Software Requirements Specification for Software Engineering: subtitle describing software

Team 19, Team BVSH
Shivya Mehta
Anindita Bornomala
Harmanpreet Singh Sagar
Vanessa Lai

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Revision History

Date	Version	Notes
Date 1	1.0	Notes
Date 2	1.1	Notes

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27 Appendix — Generative AI Use Disclosure

[ChatGPT (version 5) was used to brainstorm ideas for the template for this disclose section. —SS]

27.1 AI Tools Used

[Give the name of the AI tool(s) used and the version. —SS]

27.2 How and Where Used

[Explain what the tool(s) were used for. Examples include: brainstorming, explaining a technical concept, reviewing the document, generating or modifying text, generating or modifying diagrams, identifying errors or omissions or inconsistencies. —SS]

[For each of the uses, identify which parts of the document were affected. —SS]

[You do not need to report every interaction, but you should disclose the uses that materially contributed to your work —SS]

27.3 Verification of AI-Generated Content

[

- Describe how the team checked AI-generated or AI-assisted material before including it in the document.
- In particular, verify factual claims, technical assertions, requirements, calculations, references, and other information for which an AI system may produce incorrect or fabricated results.
- **The team is responsible for the accuracy, completeness, consistency, and appropriateness of the submitted document, regardless of whether material was generated by a human or an AI system.**

—SS]

28 Appendix — Reflection

The purpose of reflection questions is to give you a chance to assess your own learning and that of your group as a whole, and to find ways to improve in the future. Reflection is an important part of the learning process. Reflection is also an essential component of a successful software development process.

Reflections are most interesting and useful when they're honest, even if the stories they tell are imperfect. You will be marked based on your depth of thought and analysis, and not based on the content of the reflections themselves. Thus, for full marks we encourage you to answer openly and honestly and to avoid simply writing "what you think the evaluator wants to hear."

Please answer the following questions. Some questions can be answered on the team level, but where appropriate, each team member should write their own response:

1. **What went well while writing this deliverable?**

   We believe that the strongest part of the process was making useful progress by reviewing the document in small pull requests. Our team members carefully reviewed each section and highlighted duplicated descriptions, vague browser support, missing accessibility and deployment requirements, and missing or broken links. A team member also pointed out that the system-context figure, which was initially included without an explanation or legend, needed text explaining the relationships shown in the diagram. We added that explanation so the figure communicates system context rather than appearing to be included only to satisfy a documentation requirement.

2. **What pain points did you experience during this deliverable, and how did you resolve them?**

   The main difficulty was converting a broad project vision into requirements that were specific enough to verify without pretending that uncertain design decisions had already been made. At first, we described the LLM, calendar, and recommendation features in terms of capabilities, but not in terms of failure behavior or evidence of success. Choosing numerical targets was also difficult. We addressed this by reviewing comparable document-processing and scheduling systems, consulting

course requirements and project guidance, and discussing whether each target was realistic for our expected workload and resources.

Another pain point was maintaining consistency across several documents. A safety roadmap initially used identifiers that were not the labels in the Safety and Security Requirements table, and some SRS sections repeated information instead of referencing it. We resolved this through our PR reviews.

3. **How many of your requirements were inspired by speaking to your client(s) or their proxies?**

   The core requirements came from the project brief and conversations with the CAS project supervisors (Matt & Connie): outline submission, extraction of assessment information, compliance checking, shared calendar views, workload-conflict detection, and instructor control over suggested dates.

4. **Which courses have helped, or will help, your team to be successful with the capstone project?**

   Our Software Requirements course helped us distinguish stakeholder needs from functional and non-functional requirements. The course also helped us realize that statements such as "balance workloads" are not testable until they are converted into a measurable recommendation outcome. Our Software Testing course helped us define representative datasets, invalid-input cases, and verification methods. Our Database course provided the foundation for storing course and assessment information and will help us plan the structure of that information. Machine learning courses will be relevant to evaluating extraction quality, handling model uncertainty, and designing human review around LLM output.

5. **What knowledge and skills will the team collectively need to acquire?** Collectively, we need to build up our skills in a few areas to bring the different parts of the system together. This includes learning how to integrate an LLM, get consistent outputs from it, and handle issues like timeouts or unavailable services while keeping data private. We also need to figure out how to process course outline PDFs, including scanned documents, and turn the extracted information into assessment records that users can review. For scheduling, we need to

learn how to use a constraint solver like CP-SAT to find assessment conflicts, account for restricted dates, and suggest better ways to spread out workloads without automatically changing dates. On the full-stack side, we need experience with database design, API development, authentication, and connecting the frontend to the different backend components. We also need to make sure the interface is accessible and test it against WCAG guidelines. Finally, we need to learn how to deploy and configure the system properly so that someone outside our team can set it up and maintain it.

6. **How will the team acquire this knowledge and master these skills?**

   The team will develop these skills through independent research, official documentation, tutorials, and hands-on experimentation. For unfamiliar technologies or tasks, we will conduct technical spikes (which we havea already started doing when evaluating llm models and testing our PDF text extraction libraries) to test different approaches, identify limitations, and determine the best solutions before implementation. We will document the findings of each spike on GitHub Discussions so that the whole team can refer back to them throughout development. We will also share knowledge through regular meetings and code reviews, and continue learning through implementation, troubleshooting, and testing as the project progresses.